Interval Estimation — Part 2

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Topics covered

2’) Confidence Interval for the Population Mean, μ, of **Non-Normal Populations**

3. Confidence Interval for the **Proportion**, p, of a Bernoulli Population
4. Confidence Interval for the **Variance**, σ^2, and **Standard Deviation**, σ, of a Normal Population

> **Reference:** Newbold, P., Carlson, W., & Thorne, B. — *Statistics for Business and Economics*, Global Ed.

2’) CI for μ — Non-Normal Populations

Case 3 — Non-Normal Population and $n > 30$

Consider a random sample $X_1, X_2, \ldots, X_n$, $n \in \mathbb{N}$ and $n > 30$, drawn from a population X of **unknown or non-Normal distribution**.

Regardless of whether σ is known or not, the pivot statistic is always approximately $N(0,1)$ (by the Central Limit Theorem):

If σ known:

$$\underbrace{\mu}_{\text{parameter}} \longrightarrow \underbrace{\bar{X}}_{\text{point estimator}} \longrightarrow \underbrace{Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}}_{\text{pivot statistic}} \mathrel{\dot{\sim}} N(0,1)$$

If σ unknown:

$$\underbrace{\mu}_{\text{parameter}} \longrightarrow \underbrace{\bar{X}}_{\text{point estimator}} \longrightarrow \underbrace{Z = \frac{\bar{X} - \mu}{S'/\sqrt{n}}}_{\text{pivot statistic}} \mathrel{\dot{\sim}} N(0,1)$$

Case 3 — Confidence Intervals for μ

The approximate $(1-\alpha)\times 100\%$ confidence intervals for μ are:

If σ known:

$$IC_{(1-\alpha)\times 100\%}(\mu) \approx \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

If σ unknown:

$$IC_{(1-\alpha)\times 100\%}(\mu) \approx \left(\bar{x} - z_{\alpha/2}\frac{s'}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{s'}{\sqrt{n}}\right)$$

> Both intervals are **approximate**. The approximation is justified by the Central Limit Theorem when $n > 30$.

Case 3 — Application Exercise

Application Exercise — Setup

An insurance company has budgeted an average daily payout of no more than **5,000 monetary units (m.u.) per day** to cover its policyholders' claims. To validate this estimate, 100 days were randomly selected from the company's historical records, yielding:

$$\bar{x} = 5{,}625 \text{ m.u.} \qquad s' = 2{,}500 \text{ m.u.}$$

Find a 95% CI for the mean daily payout and determine whether 5,000 m.u./day is sufficient.

Application Exercise — Solution

Population distribution: **unknown**; σ: **unknown**.

Sample: $n = 100$, $\bar{x} = 5{,}625$, $s' = 2{,}500$.

Pivot statistic: $Z = \dfrac{\bar{X} - \mu}{S'/\sqrt{n}} \,\dot{\sim}\, N(0,1)$

$$1 - \alpha = 0.95 \;\Leftrightarrow\; \alpha = 0.05 \;\Rightarrow\; z_{\alpha/2} = 1.96$$

$$IC_{95\%}(\mu) \approx \left(5{,}625 - 1.96 \times \frac{2{,}500}{\sqrt{100}} \;;\; 5{,}625 + 1.96 \times \frac{2{,}500}{\sqrt{100}}\right)$$

$$\boxed{IC_{95\%}(\mu) \approx (5{,}135 \;;\; 6{,}115)}$$

Application Exercise — Conclusion

$$IC_{95\%}(\mu) \approx (5{,}135\ ;\ 6{,}115)$$

Conclusion

$$5{,}000 \notin IC_{95\%}(\mu)$$

The value 5,000 m.u. lies **below** the entire confidence interval. With 95% confidence, the true mean daily payout **exceeds** 5,000 m.u./day. The company's budget is likely **insufficient**.

3) CI for the Proportion p

CI for the Proportion — Setup

Consider a random sample $X_1, X_2, \ldots, X_n$, $n \in \mathbb{N}$, $n > 30$, drawn from a **Bernoulli** population:

$$X \sim \text{Bernoulli}(p), \quad 0 < p < 1$$

(equivalently, $X \sim \text{Binomial}(1, p)$)

$$\underbrace{p}_{\text{parameter}} \longrightarrow \underbrace{\frac{Y}{n}}_{\substack{\text{point}\\ \text{estimator}}} \longrightarrow \underbrace{Z = \frac{\hat{p} - p}{\sqrt{\hat{p}\hat{q}/n}}}_{\text{pivot statistic}} \mathrel{\dot{\sim}} N(0,1)$$

where $Y = \sum_{i=1}^{n} X_i$, $\hat{p} = \bar{X} = \dfrac{Y}{n}$, and $\hat{q} = 1 - \hat{p}$.

CI for the Proportion — Formula

The approximate $(1-\alpha)\times 100\%$ **confidence interval for** p is:

$$\boxed{IC_{(1-\alpha)\times 100\%}(p) \approx \left(\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}\,\hat{q}}{n}},\ \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}\,\hat{q}}{n}}\right)}$$

Conditions for validity: $n > 30$, and ideally $n\hat{p} \geq 5$ and $n\hat{q} \geq 5$, so that the Normal approximation to the Binomial is reliable.

Application Exercise — Proportion

Application Exercise — Setup

In a given district, **840 out of 2,000 voters** surveyed in a poll declared they would vote for Party A.

Construct a 95% confidence interval for the proportion of voters supporting Party A.

Population: $X \sim \mathrm{Bernoulli}(p)$, where p = proportion of Party A voters.

Sample: $n = 2{,}000$; $x = 840$ (observed Party A voters in the sample).

$$\hat{p} = \frac{x}{n} = \frac{840}{2{,}000} = 0.42 \qquad \hat{q} = 1 - \hat{p} = 0.58$$

Application Exercise — Solution

Pivot statistic: $Z = \dfrac{\hat{p} - p}{\sqrt{\hat{p}\hat{q}/n}} \dot{\sim} N(0,1)$

$$1 - \alpha = 0.95 \;\Leftrightarrow\; \alpha = 0.05 \;\Rightarrow\; z_{\alpha/2} = 1.96$$

$$IC_{95\%}(p) \approx \left(0.42 - 1.96\sqrt{\frac{0.42 \times 0.58}{2{,}000}}\;;\; 0.42 + 1.96\sqrt{\frac{0.42 \times 0.58}{2{,}000}}\right)$$

95% Confidence Interval for p

$IC_{95\%}(p) \approx (0.3983\;;\;0.4416)$

With 95% confidence, the proportion of Party A voters in the district is between **39.83%** and **44.16%**.

4) CI for σ^2 and σ

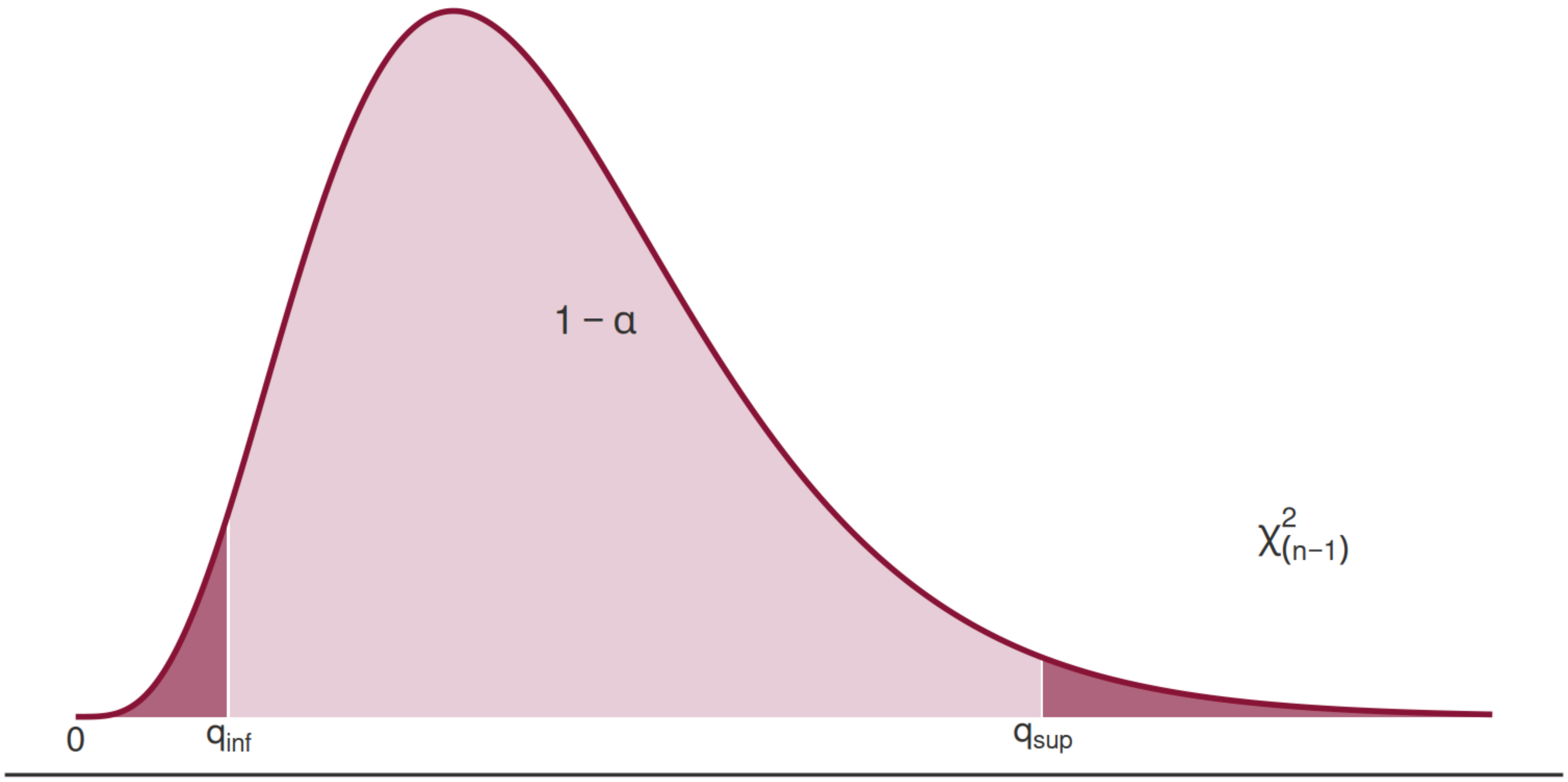
CI for the Variance — Setup

Consider a random sample $X_1, X_2, \ldots, X_n$, $n \in \mathbb{N}$, drawn from a **Normal** population $X \sim N(\mu, \sigma)$, with both μ and σ **unknown**.

$$\underbrace{\sigma^2}_{\text{parameter}} \longrightarrow \underbrace{S'^{\,2}}_{\substack{\text{point}\\ \text{estimator}}} \longrightarrow \underbrace{Q = \frac{(n-1)S'^{\,2}}{\sigma^2}}_{\text{pivot statistic}} \sim \chi^2_{(n-1)}$$

The pivot statistic Q follows **exactly** a chi-squared distribution with $n-1$ degrees of freedom.

CI for σ^2 — Probability Statement



$$P(Q < q_{\text{inf}}) = P(Q > q_{\text{sup}}) = \frac{\alpha}{2}$$

CI for σ^2 and σ — Formulas

The $(1-\alpha)\times 100\%$ **confidence interval for** σ^2 is:

$$\boxed{IC_{(1-\alpha)\times 100\%}(\sigma^2) = \left(\frac{(n-1)\,s'^{\,2}}{q_{\text{sup}}}\,,\;\frac{(n-1)\,s'^{\,2}}{q_{\text{inf}}}\right)}$$

If a CI for σ is required, take the square root of each bound:

$$\boxed{IC_{(1-\alpha)\times 100\%}(\sigma) = \left(\sqrt{\frac{(n-1)\,s'^{\,2}}{q_{\text{sup}}}}\,,\;\sqrt{\frac{(n-1)\,s'^{\,2}}{q_{\text{inf}}}}\right)}$$

where q_{inf} and q_{sup} are the lower and upper critical values of the $\chi^2_{(n-1)}$ distribution.

Application Exercise — Variance

Application Exercise — Setup

Based on a random sample of $n = 16$ observations drawn from a **Normal population**, a corrected sample standard deviation of $s' = 3.872$ was obtained.

Construct a 95% confidence interval for the population variance.

Population: $X \sim N(\mu, \sigma)$, both parameters unknown.

Sample: $n = 16$, $s' = 3.872$.

Pivot statistic:

$$Q = \frac{(n-1)\, S'^{\,2}}{\sigma^2} \sim \chi^2_{(n-1)} \equiv \chi^2_{(15)}$$

Application Exercise — Finding the Critical Values

We need $q_{\inf}$ and $q_{\sup}$ from the $\chi^2_{(15)}$ distribution (Table 6, row 15):

$$1-\alpha = 0.95 \;\Leftrightarrow\; \alpha = 0.05 \;\Rightarrow\; \frac{\alpha}{2} = 0.025$$

$$P(Q > q_{\sup}) = \frac{\alpha}{2} = 0.025 \quad \underset{\substack{\text{Table 6} \\ \text{row 15, } \varepsilon=0.025}}{\Longrightarrow} \quad q_{\sup} = 27.488$$

$$P(Q < q_{\inf}) = \frac{\alpha}{2} \;\Leftrightarrow\; P(Q > q_{\inf}) = 1 - \frac{\alpha}{2} = 0.975 \quad \underset{\substack{\text{Table 6} \\ \text{row 15, } \varepsilon=0.975}}{\Longrightarrow} \quad q_{\inf} = 6.262$$

Application Exercise — Result for σ^2

$$IC_{95\%}(\sigma^2) = \left(\frac{(16-1)\times 3.872^2}{27.488} \; ; \; \frac{(16-1)\times 3.872^2}{6.262} \right)$$

95% Confidence Interval for σ^2

$$IC_{95\%}(\sigma^2) = (8.181 \; ; \; 35.913)$$

Application Exercise — Result for σ

If a confidence interval for σ is required, take the square root of each bound:

$$IC_{95\%}(\sigma) = \left(\sqrt{\frac{(16-1) \times 3.872^2}{27.488}} \; ; \; \sqrt{\frac{(16-1) \times 3.872^2}{6.262}} \right)$$

95% Confidence Interval for σ

$$IC_{95\%}(\sigma) = (2.860 \; ; \; 5.993)$$

Summary — All Cases

Case	Parameter	Population	n	Pivot	CI type
1	μ	Normal, σ known	any	$Z \sim N(0,1)$	Exact
2	μ	Normal, σ unknown	≤ 30	$T \sim t_{(n-1)}$	Exact
2*	μ	Normal, σ unknown	> 30	$Z \dot{\sim} N(0,1)$	Approx.
3	μ	Any, σ known	> 30	$Z \dot{\sim} N(0,1)$	Approx.
3*	μ	Any, σ unknown	> 30	$Z \dot{\sim} N(0,1)$	Approx.
—	p	Bernoulli	> 30	$Z \dot{\sim} N(0,1)$	Approx.
—	σ^2	Normal	any	$Q \sim \chi^2_{(n-1)}$	Exact